

DutchSemCor

**methodology
approaches
tools**

SemCor: a Semantic Concordance

- “...a **textual corpus** and a **lexicon** so **combined** that every substantive word in the text is linked to its appropriate sense in the lexicon.”

“...it can be viewed either as a corpus in which words have been tagged syntactically and semantically, or as a lexicon in which example sentences can be found for many definitions.”

(Miller et al. 1993)

SemCor: a Semantic Concordance

SemCor (All-words corpus)

- WordNet team (Princeton University)
- **Brown Corpus + WordNet 1.6**
- 186 texts all content words (N, V, Adj, Adv)
160 texts only verbs
- 233K tokens

SemCor: a Semantic Concordance

DutchSemCor (Lexical-sample corpus)

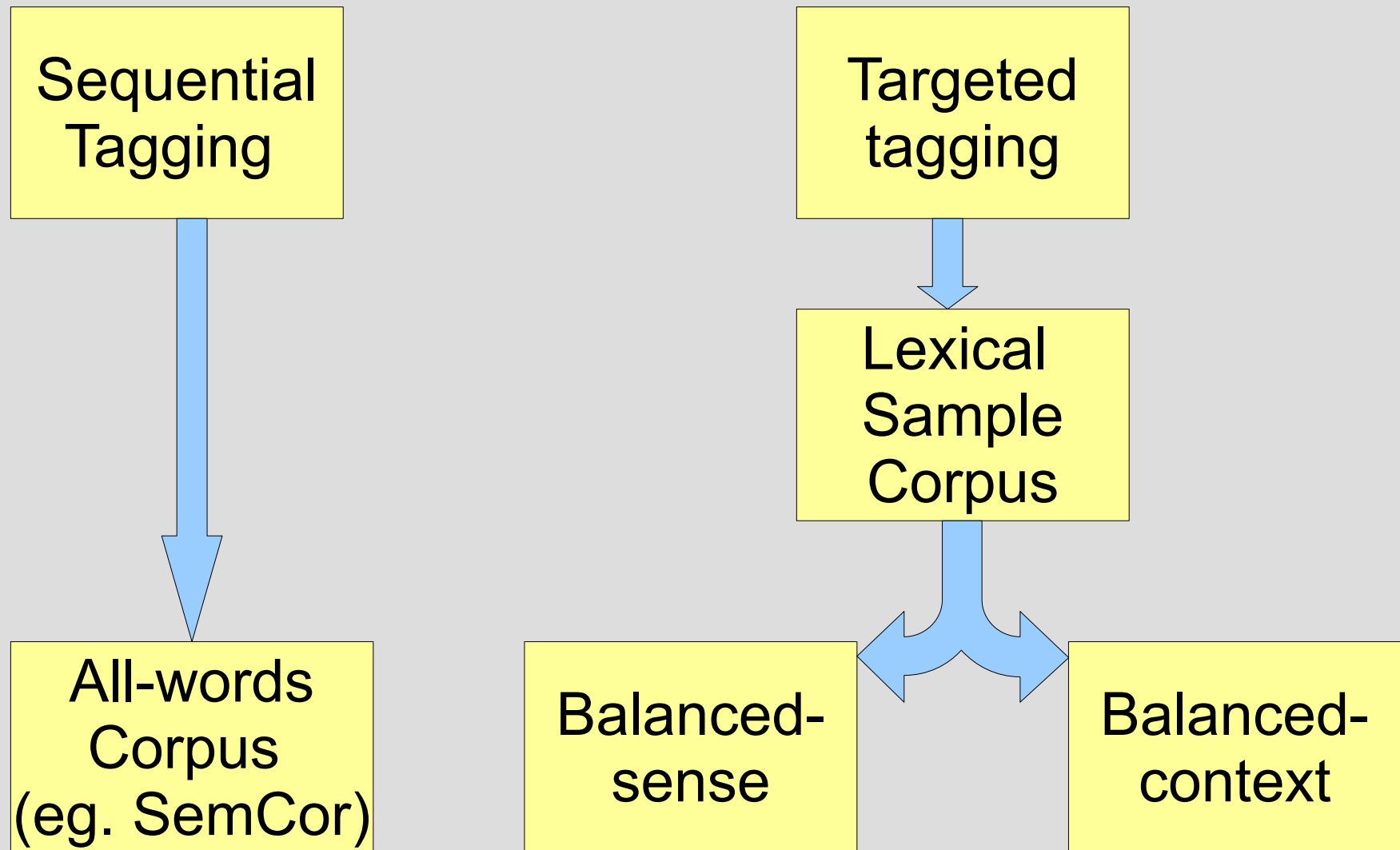
- VU-UvA-UvT-UvN (NWO)
- **SoNaR 500M & CGN 1M & Snippets + Cornetto**

SemCor: a Semantic Concordance

DutchSemCor (Lexical-sample corpus)

- VU-UvA-UvT-UvN (NWO)
- **SoNaR 500M & CGN 1M & Snippets + Cornetto**
- **Anticipated circa 1 million tokens**
 - **3.000** most polysemous & frequent Dutch words
 - **100** tokens per sense (25 manually/ 75 manually+automatically)
 - **3,4** senses per lemma

Creating a sense-tagged corpus



The ideal sense-tagged corpus?

- All-words (e.g. whole SoNaR tagged)
 - :) info on sense distributions
 - : (8 annotators full-time 123 years!!!
 - : (Not all senses would be (equally) represented
 - ? Diversity of contexts depends on corpus used

The ideal sense-tagged corpus?

- **Balanced-sense (part of SoNaR tagged + extended)**
 - :) provide examples for all senses of selected lemmas**
 - : (info on sense distributions**
 - : (might not be diverse in contexts**

The ideal sense-tagged corpus?

- **Balanced-context (part of SoNaR tagged + filtered)**
 - :) provide diversity in contexts**
 - : (info on sense distributions**
 - : (might not contain all senses**

The ideal sense-tagged corpus?

	sense- distribution	sense- coverage	context- diversity
All-words	:)	: (	?
Balanced- sense	: (	:)	: (
Balanced- context	: (	: (	:)

The ideal sense-tagged corpus?

	sense- distribution	sense- coverage	context- diversity
All-words	:)	: (	?
Balanced- sense	: (	:)	: (
Balanced- context	: (	: (	:)

The ideal sense-tagged corpus?

	sense- distribution	sense- coverage	context- diversity
All-words	:)	: (	?
Balanced- sense	: (	:)	: (
Balanced- context	: (	: (	:)

The ideal sense-tagged corpus?

	sense- distribution	sense- coverage	context- diversity
All-words	:)	: (	?
Balanced- sense	: (	:)	: (
Balanced- context	: (	: (	:)

1st phase: Balanced-sense corpus

- 25 representative examples for each sense
- IA > 80%
- Overlapping annotations
- Filter words
- Extending SoNaR with web-snippets
- Tools: SAT, SNIPPET-tool, Loganalyser

2nd phase: Balanced-context corpus

- Active Learning
- add examples for low performing words until we reach accuracy of 80% or no progress
- High confidence, high distance
- Best results when training on agreed tokens

3rd phase: Automatic sense-tagging remaining corpus

- Training WSD-systems with human annotated data
- Apply WSD to rest of the full corpus
- Circa 48M tokens

Additional subcorpora

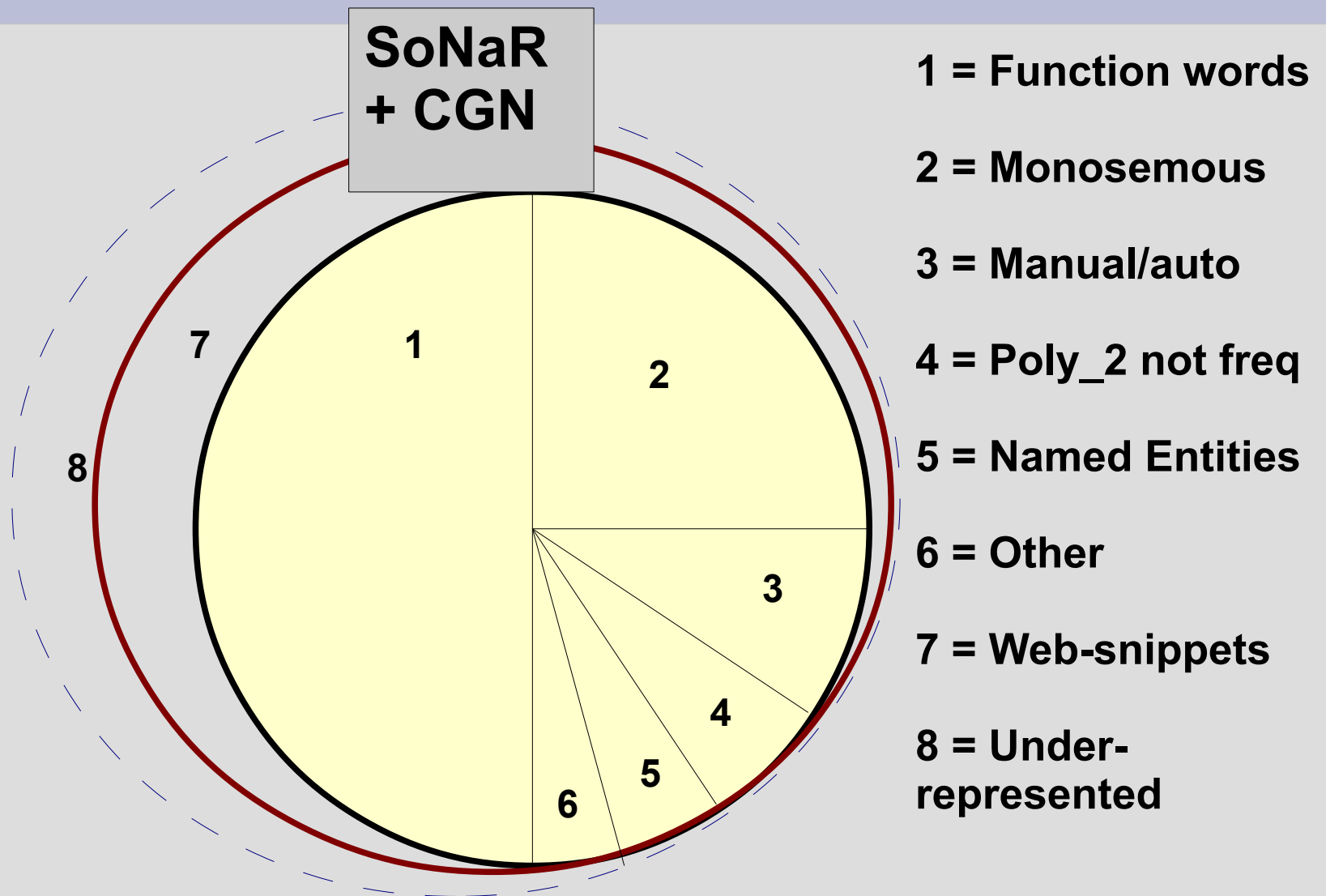
- **Random evaluation corpus**
 - + Between 90 and 100 (not with acc=100)
 - + Between 80 and 90
 - + Between 70 and 80
 - + Between 60 and 70
 - Number of tokens = 5.200
 - Number of lemmas = 52

Additional subcorpora

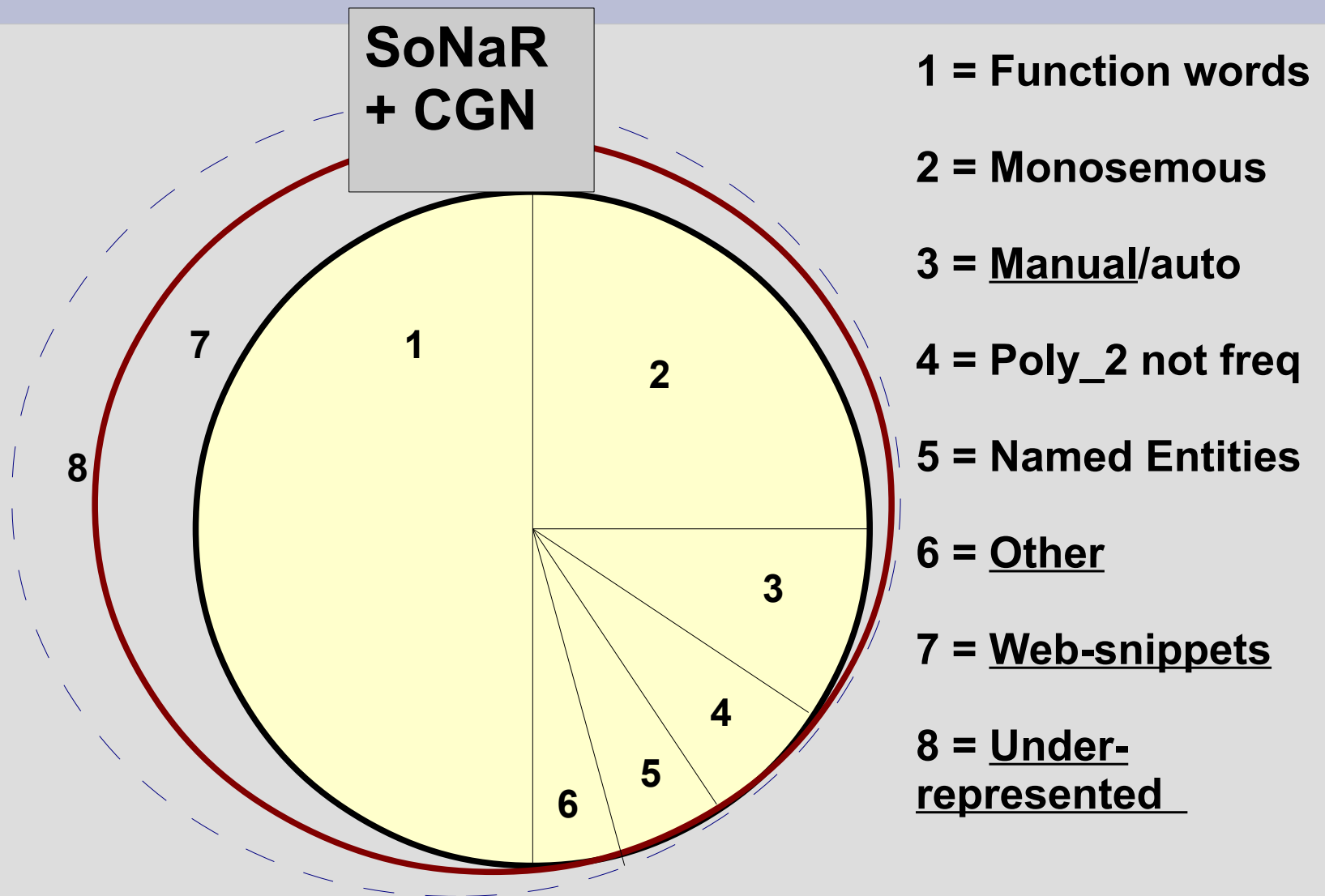
- **All-words corpus**

- Nmr tokens = 23.907
- Nmr lemmas = 1.527
- Medical journals, manuals, newspapers, periodicals, magazines, reports/ surveys, websites, wikipedia
- Part of LASSY corpus (syntactically annotated)

Coverage

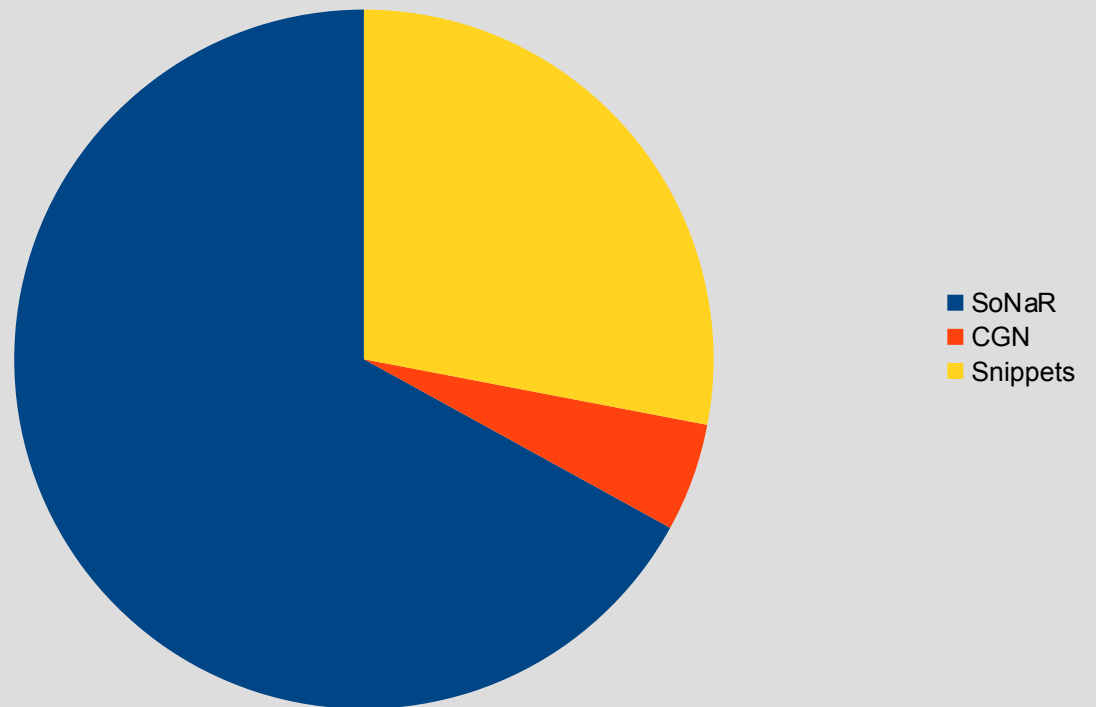


Coverage



Manual annotations (1st & 2nd phase)

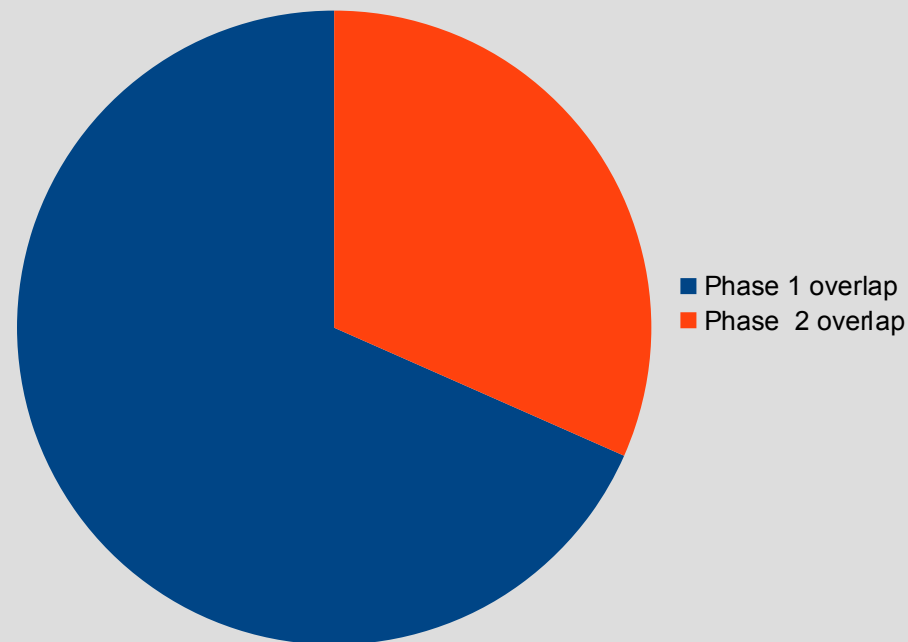
SoNaR 67%
CGN 5%
Snippets 28%



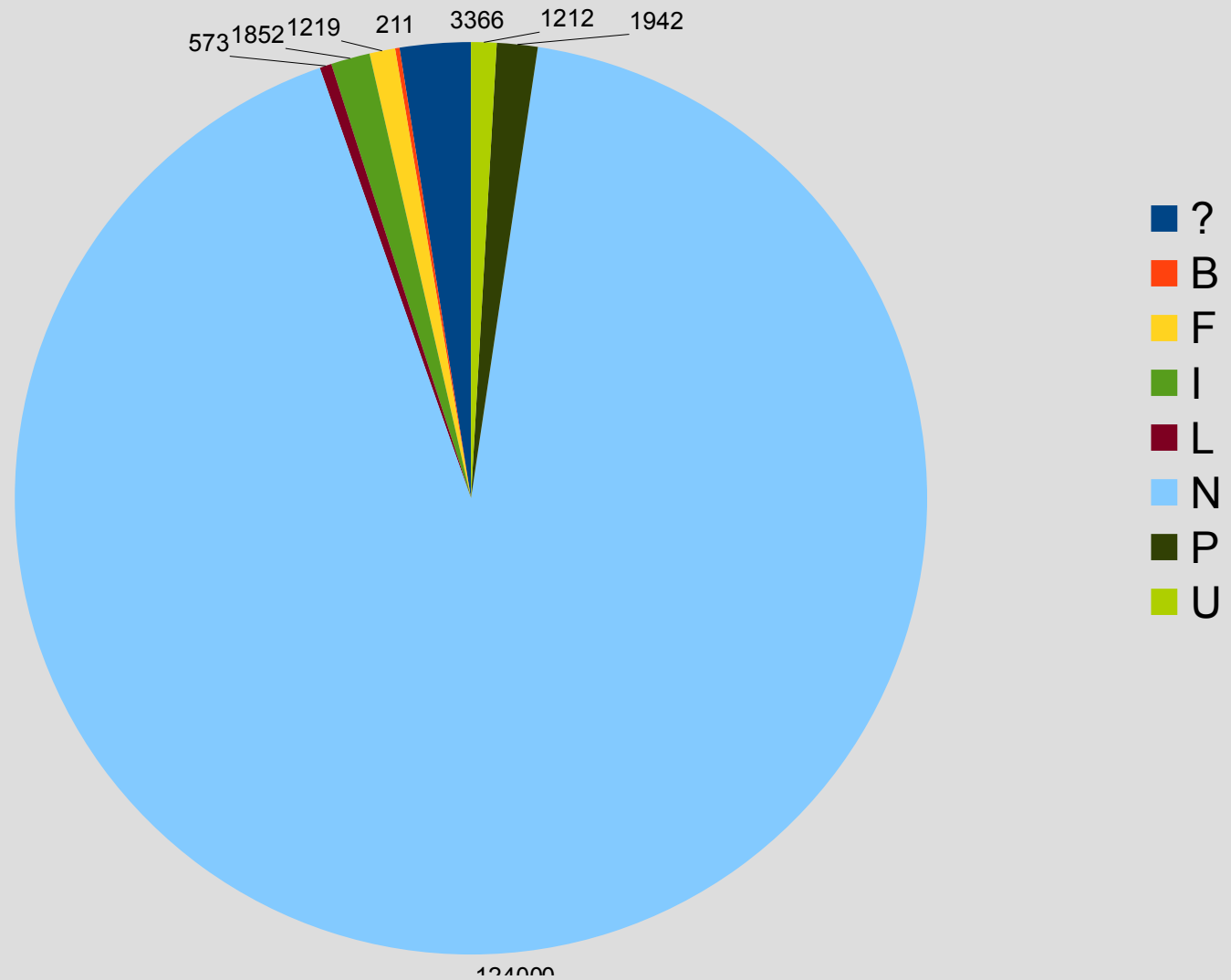
Manual annotations (1st & 2nd phase)

Total tokens 1 – 338K
Overlap tokens 1 – 286K
IA = 92%

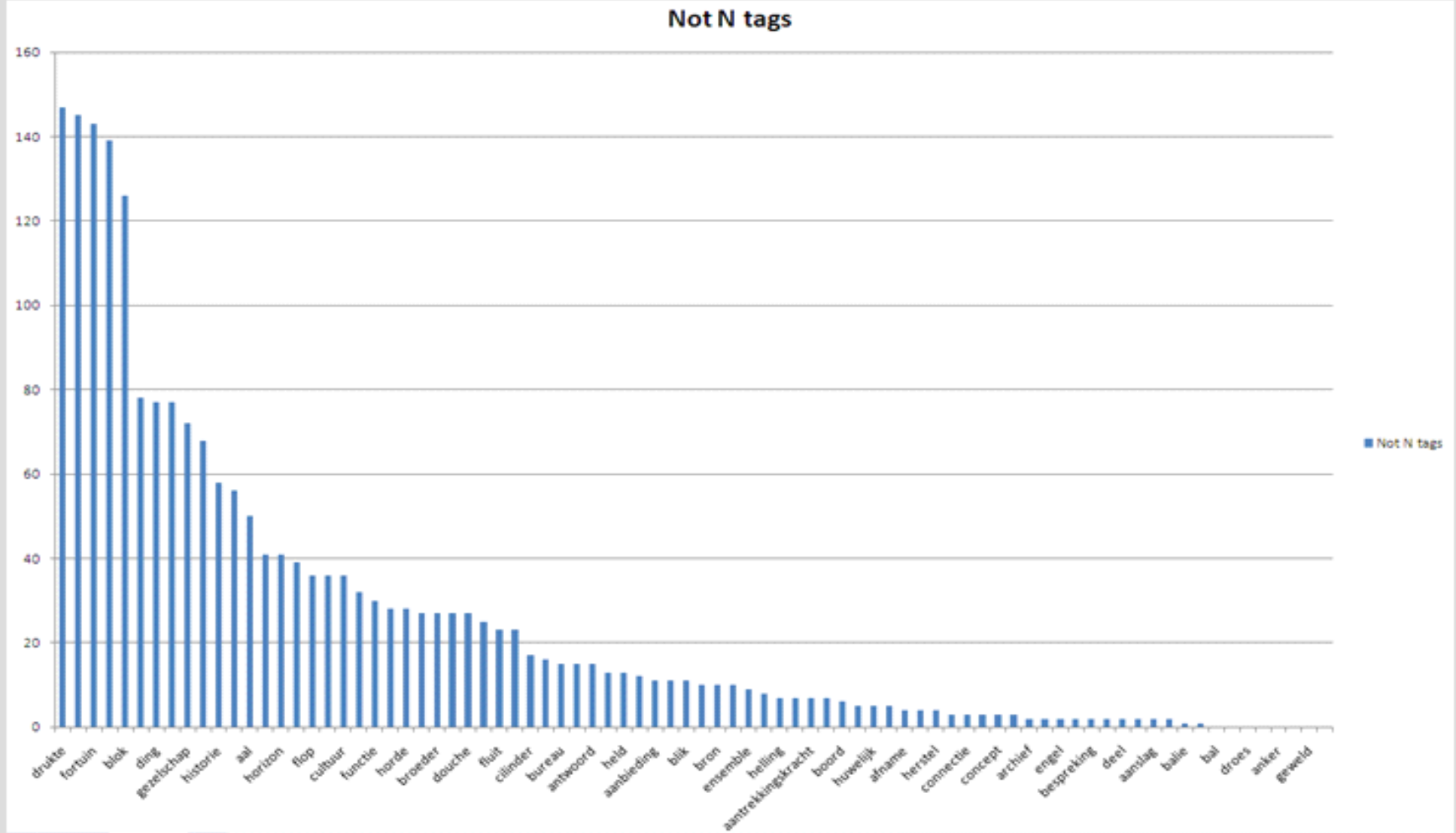
Total tokens 2 – 144K
Overlap tokens 2 – 132K



Other = 8%



Other = 8%



Other = 8%

?

*La **parade** du Dragon Récemment les Chinois de Paris ont fêté avec un enthousiasme exceptionnel*

F

*Hij is een breed-spectrum musicus, hij is een **kameleon***

U

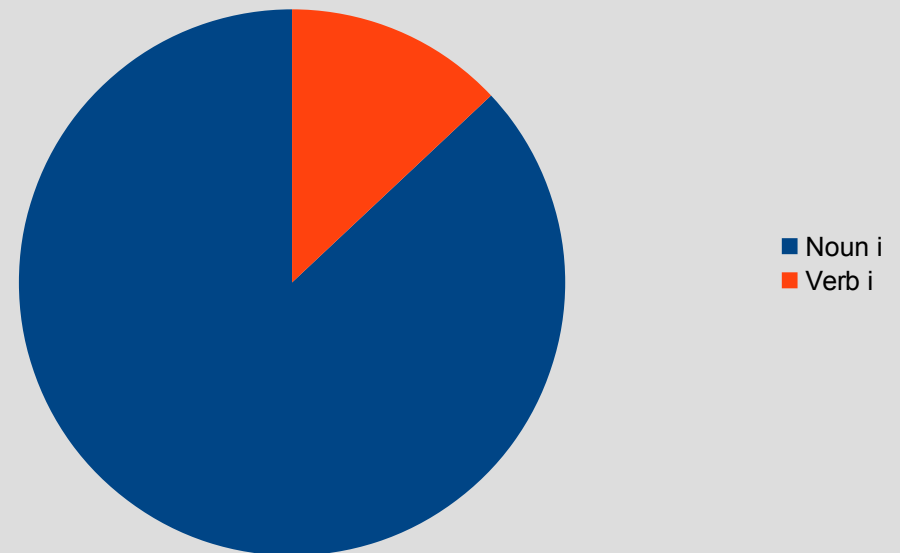
*Met de **reductie** van Groningen wordt de capitulatie voor het leger van Maurits van de stad Groningen op 22 juli 1594 aangeduid*

Other = 8%

Total idioms SoNaR: 118.978

103.344 nouns -> 87 %

15.642 verbs -> 13 %



Under-represented

Coverage of senses with 25 examples: 71%

Coverage of annotations for words: 80%

kandidaat:4

drager laagste academische graad

crisis:2

kritieke fase in een ziekte

bloem:2

top (“de bloem der natie”)

Conclusion

**Lots of interesting data
for the future:**

<http://www2.let.vu.nl/oz/cltl/dutchsemcor/>

...Coffee!